

WASP-43b

R-0609

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180321 · created 2026-07-25T07:09:34+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458198.693 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.024
Transit depth (Rp/Rs)^2	1.87 +/- 0.67 [%]
Orbital Inclination (inc)	86.46 +/- 1.98
Airmass coefficient 1 (a1)	1.007 +/- 0.019
Airmass coefficient 2 (a2)	-0.004 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#3 - [374, 30]
Optimal Aperture	3.63
Optimal Annulus	9.12
Transit Duration (day)	0.058 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.137 ± 0.024 vs 0.1594 (deviation 0.93σ)
Duration fit vs published	0.058 d vs 0.0512 d (deviation 2.33σ)
Mid-time O–C	3.5 min
Depth SNR	5.7
Residual scatter	1.93 %

Light curve

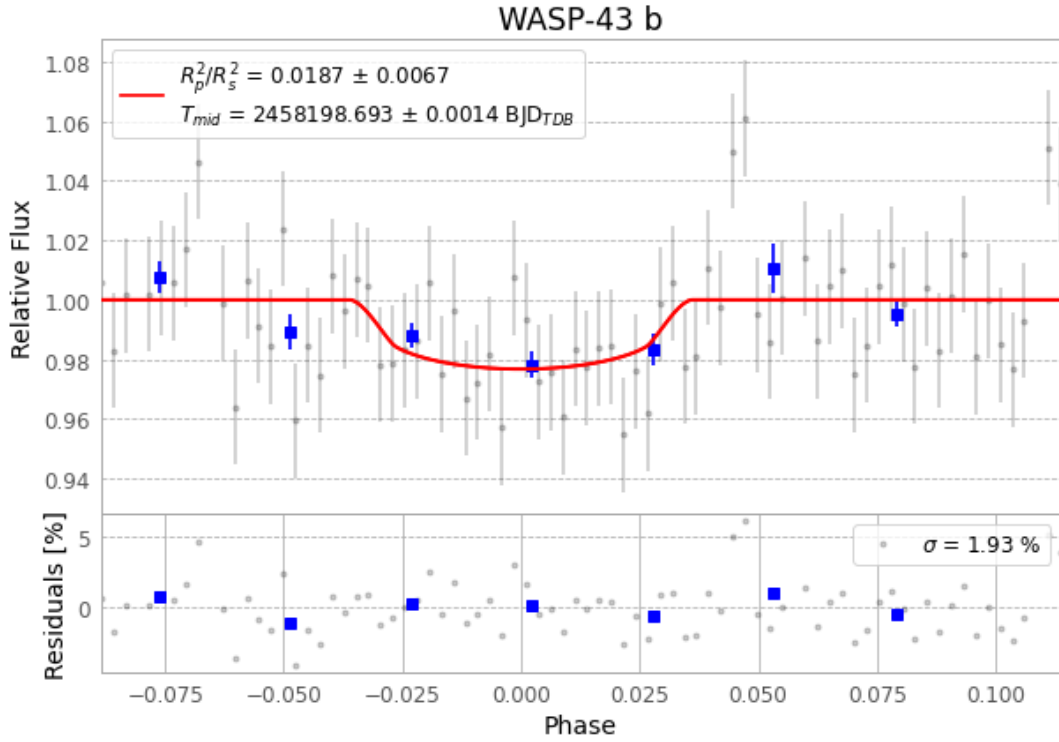


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-20.png (SHA-256 13fe0174647e2b12...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [461, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb7ce8fe1850f167...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

80 FITS frames (53 MB), WASP-43180321024512.FITS → WASP-43180321064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180321.log (e5e6ea558a81...), FinalLightCurve_WASP-43 b_2018-03-20.png (13fe0174647e...), FinalLightCurve_WASP-43 b_2018-03-20.csv (a91a754933b0...)

Compute resources

Wall time 355.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 07:09Z.